

V4.2 Toy Model 3E

Gauge Phase and Electromagnetic Coupling as Local Phase-Connection Structure

Rev A - Exploratory Mathematical Compatibility Report

Date: 2026.05.25 | Continuation after Toy Model 3D

Status: Toy Model 3E tests whether electromagnetic minimal coupling can be represented as the connection required to compare local hidden phase orientations across space and time. It does not derive Maxwell equations or the physical origin of electric charge.

1. Purpose

Toy Model 3D represented scalar potentials as local hidden phase-rotation-rate shifts. Toy Model 3E asks the next question: if the phase reference itself can vary from point to point, what mathematical object is needed to compare phase at neighboring locations?

The standard answer is a gauge connection. In quantum mechanics with electromagnetism, this connection appears as the vector potential A and scalar potential Φ . The target here is structural compatibility, not derivation of electromagnetism from first principles.

2. Local phase transformation

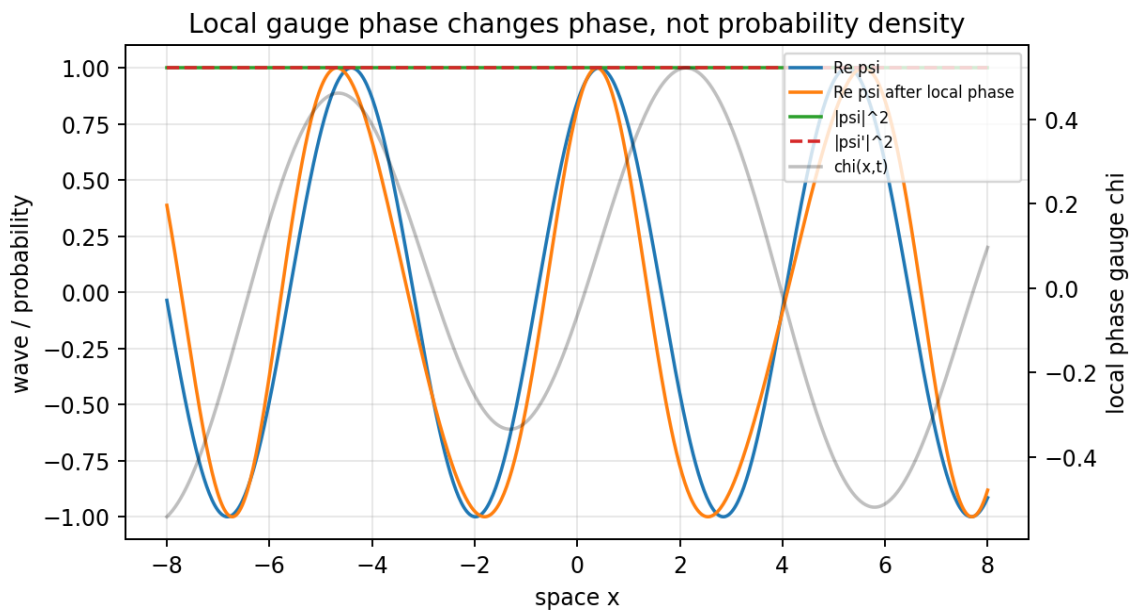
Apply a local phase transformation:

$$\psi' = \exp(i q \chi(x,t) / \hbar) \psi$$

Term reminders:

- ψ is the quantum amplitude.
- q is electric charge.
- $\chi(x,t)$ is an arbitrary local phase-reference function.
- $\hbar$ is the reduced Planck constant.
- local means χ can be different at different positions and times.

In 6D language, this means the hidden transverse phase plane has a different local reference orientation at each point in 3S space and 1T time.



3. Why the ordinary derivative breaks

The ordinary spatial derivative of the transformed state is:

$$d\psi'/dx = \exp(i q \chi/\hbar) [d\psi/dx + i(q/\hbar)(d\chi/dx) \psi]$$

The extra $d\chi/dx$ term means the ordinary derivative does not transform in the same simple way as ψ . This is the broken-derivative problem.

Plain meaning: if neighboring hidden phase planes use different local reference angles, ordinary subtraction between neighboring amplitudes is not meaningful until the reference frames are connected.

4. Covariant connection

Introduce the gauge potentials and covariant derivatives:

$$\begin{aligned} D_x &= d/dx - i(q/\hbar) A \\ D_t &= d/dt + i(q/\hbar) \Phi \end{aligned}$$

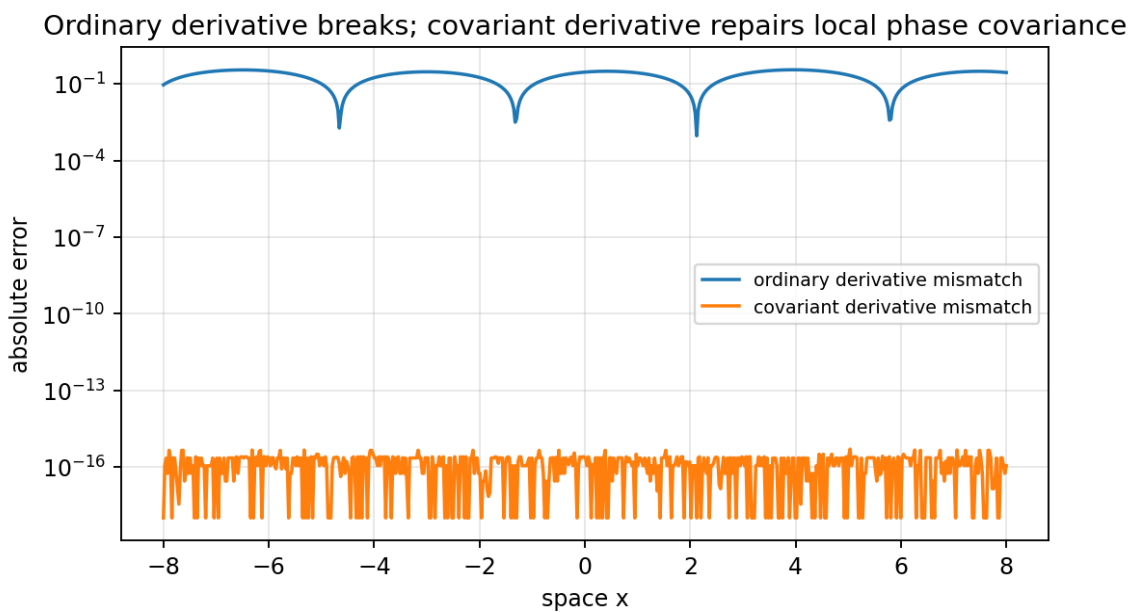
Use the gauge transformation rules:

$$\begin{aligned} A' &= A + d\chi/dx \\ \Phi' &= \Phi - d\chi/dt \end{aligned}$$

Then the covariant derivatives transform cleanly:

$$\begin{aligned} D'_x \psi' &= \exp(i q \chi/\hbar) D_x \psi \\ D'_t \psi' &= \exp(i q \chi/\hbar) D_t \psi \end{aligned}$$

This is the core result: A and Phi act as the connection that allows different local phase planes to be compared consistently.



5. Electromagnetic minimal coupling structure

The covariant Schrodinger form is:

$$i \hbar D_t \psi = -(\hbar^2/2m) D_x^2 \psi$$

Expanding D_t gives:

$$i \hbar d\psi/dt = [-(\hbar^2/2m) D_x^2 + q \Phi] \psi$$

Since:

$$-i \hbar D_x = -i \hbar d/dx - q A$$

the Hamiltonian becomes the standard minimal-coupling form:

$$i \hbar d\psi/dt = [(1/2m)(-i \hbar d/dx - q A)^2 + q \Phi] \psi$$

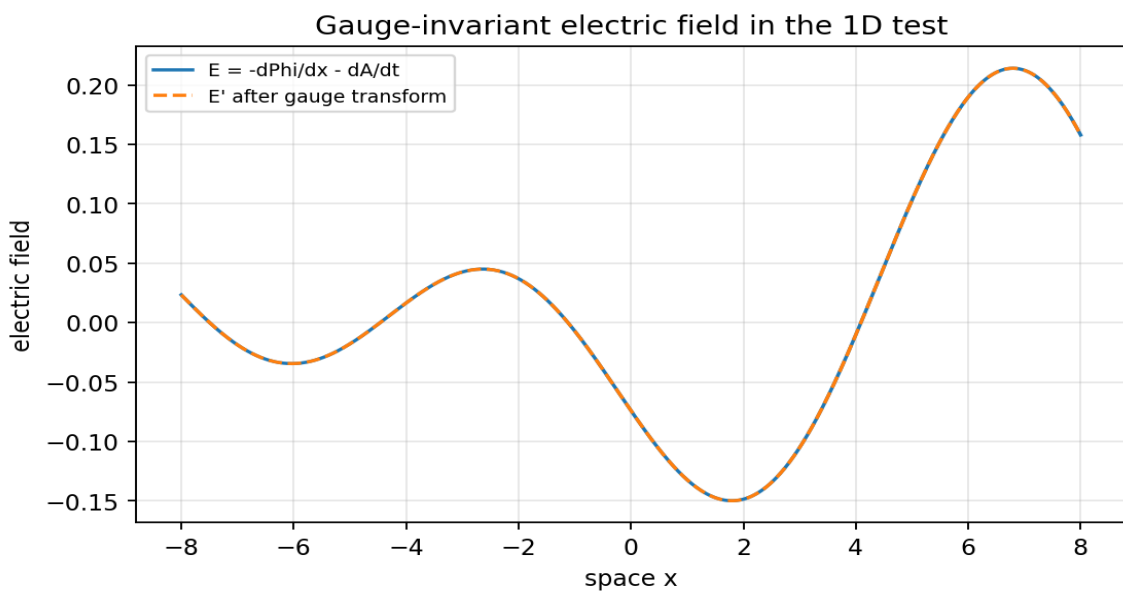
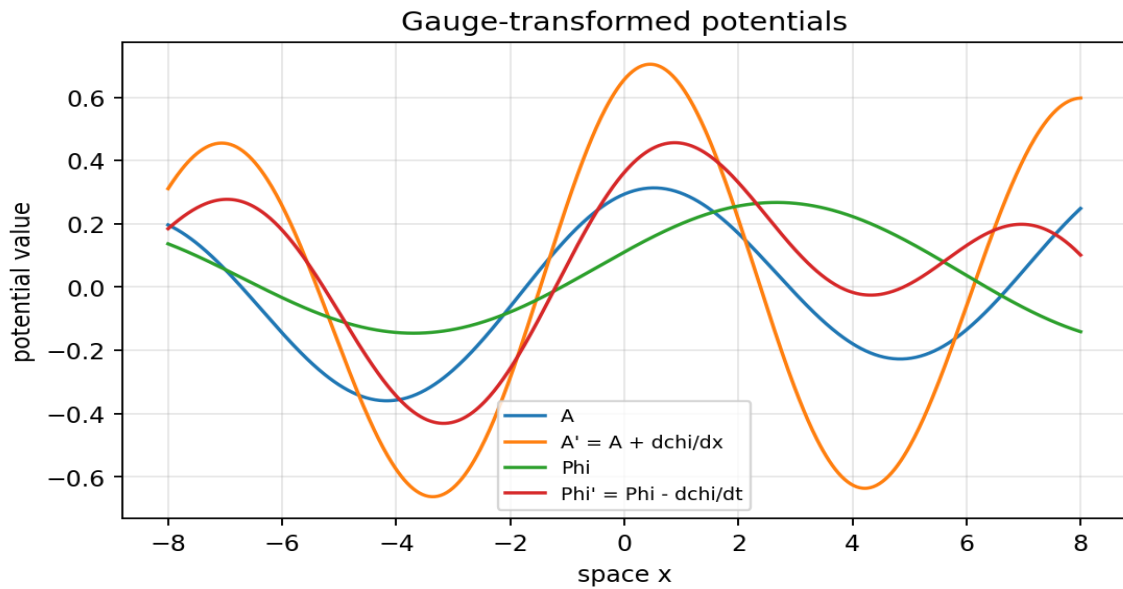
This is the usual electromagnetic coupling structure in one spatial dimension. In three dimensions, d/dx becomes the gradient operator and A becomes a vector potential.

6. Gauge-invariant fields

The potentials themselves can change under a gauge transformation, but the physical electric field remains unchanged. In the one-dimensional test:

$$E = -d\Phi/dx - dA/dt$$

Using A' and Φ' gives the same E . This is why A and Φ are connection variables, while E is gauge-invariant.



7. Numerical checks

Check	Result
Random samples	10000
Mean ordinary derivative mismatch	2.063e-01
Max ordinary derivative mismatch	3.576e-01
Max covariant spatial derivative error	1.006e-15
Max covariant time derivative error	9.930e-16
Max covariant second-derivative error	3.794e-15
Max covariant Schrodinger residual error	1.831e-15
Max probability-density gauge error	4.441e-16
Max electric-field gauge error	5.551e-17

8. Interpretation inside the 6D Projection Framework

Toy Model 3E supports the following limited interpretation:

- local hidden phase planes vary across $3S + 1T$
 - > ordinary derivatives fail
 - > connection fields are required
 - > A connects spatial phase orientation
 - > Φ connects temporal phase orientation
 - > minimal electromagnetic coupling structure appears

In this interpretation, gauge potentials are not arbitrary decoration. They are the bookkeeping needed to compare local hidden temporal phase orientation consistently across the observer-accessible projection.

9. What Toy Model 3E achieves

- It shows why local phase freedom breaks ordinary derivatives.
- It introduces covariant derivatives as the repair mechanism.
- It recovers the standard minimal-coupling structure for electromagnetic potentials.
- It verifies gauge covariance of D_x , D_t , D_x^2 , and the covariant Schrodinger residual.
- It verifies probability-density invariance under local phase transformation.
- It verifies electric-field invariance under the one-dimensional gauge test.

10. What Toy Model 3E does not achieve

- It does not derive Maxwell equations.
- It does not derive electric charge q .
- It does not prove the physical existence of 3T.
- It does not derive the electromagnetic field from matter sources.
- It does not derive relativistic quantum electrodynamics.
- It does not derive non-Abelian gauge fields or the Standard Model.

11. Close-out statement for 3E

Toy Model 3E successfully formalizes electromagnetic minimal coupling as a local phase-connection structure. A local phase transformation $\psi \rightarrow \exp(i q \chi/\hbar) \psi$ breaks ordinary derivatives because neighboring amplitudes no longer share the same phase reference. Introducing A and Φ as connection fields restores covariance through $D_x = d/dx - i(q/\hbar)A$ and $D_t = d/dt + i(q/\hbar)\Phi$. The resulting Schrodinger structure matches minimal electromagnetic coupling. This is a structural compatibility result, not a derivation of Maxwell equations, electric charge, or the Standard Model.

12. Recommended next chapter

The next logical chapter is Toy Model 3F: Relativistic Phase Constraint and Klein-Gordon/Dirac Bridge.

1. Test whether the energy-momentum relation $E^2 = p^2 c^2 + m^2 c^4$ can be represented through phase geometry.
2. Recover the Klein-Gordon structural form first.
3. Then test whether first-order factorization naturally leads toward Dirac structure.
4. Maintain the guardrail: structural compatibility only, not full particle physics.
5. Identify where spin, antiparticles, and Lorentz symmetry enter the 6D projection framework.

13. Rebind prompt for continuation

We are resuming after Toy Model 3E.

3E tested gauge phase and electromagnetic coupling as local phase-connection structure.

Local phase transformation:

$$\psi' = \exp(i q \chi(x,t)/\hbar) \psi$$

Ordinary derivative breaks because:

$$d\psi'/dx = \exp(i q \chi/\hbar) [d\psi/dx + i(q/\hbar)(d\chi/dx)\psi]$$

Introduce connection fields:

$$D_x = d/dx - i(q/\hbar) A$$

$$D_t = d/dt + i(q/\hbar) \Phi$$

Gauge transformations:

$$A' = A + d\chi/dx$$

$$\Phi' = \Phi - d\chi/dt$$

Then:

$$D'_x \psi' = \exp(i q \chi/\hbar) D_x \psi$$

$$D'_t \psi' = \exp(i q \chi/\hbar) D_t \psi$$

Covariant Schrodinger structure:

$$i \hbar D_t \psi = -(\hbar^2/2m) D_x^2 \psi$$

Expanded:

$$i \hbar \frac{d\psi}{dt} = \left[\frac{1}{2m} (-i \hbar \frac{d}{dx} - qA)^2 + q\Phi \right] \psi$$

Result:

Electromagnetic minimal coupling is structurally equivalent to maintaining a consistent local phase connection across space and time.

Guardrail:

3E does not derive Maxwell equations, charge q , QED, the Standard Model, or physical 3T existence.

Next recommended chapter:

Toy Model 3F - Relativistic Phase Constraint and Klein-Gordon/Dirac Bridge.